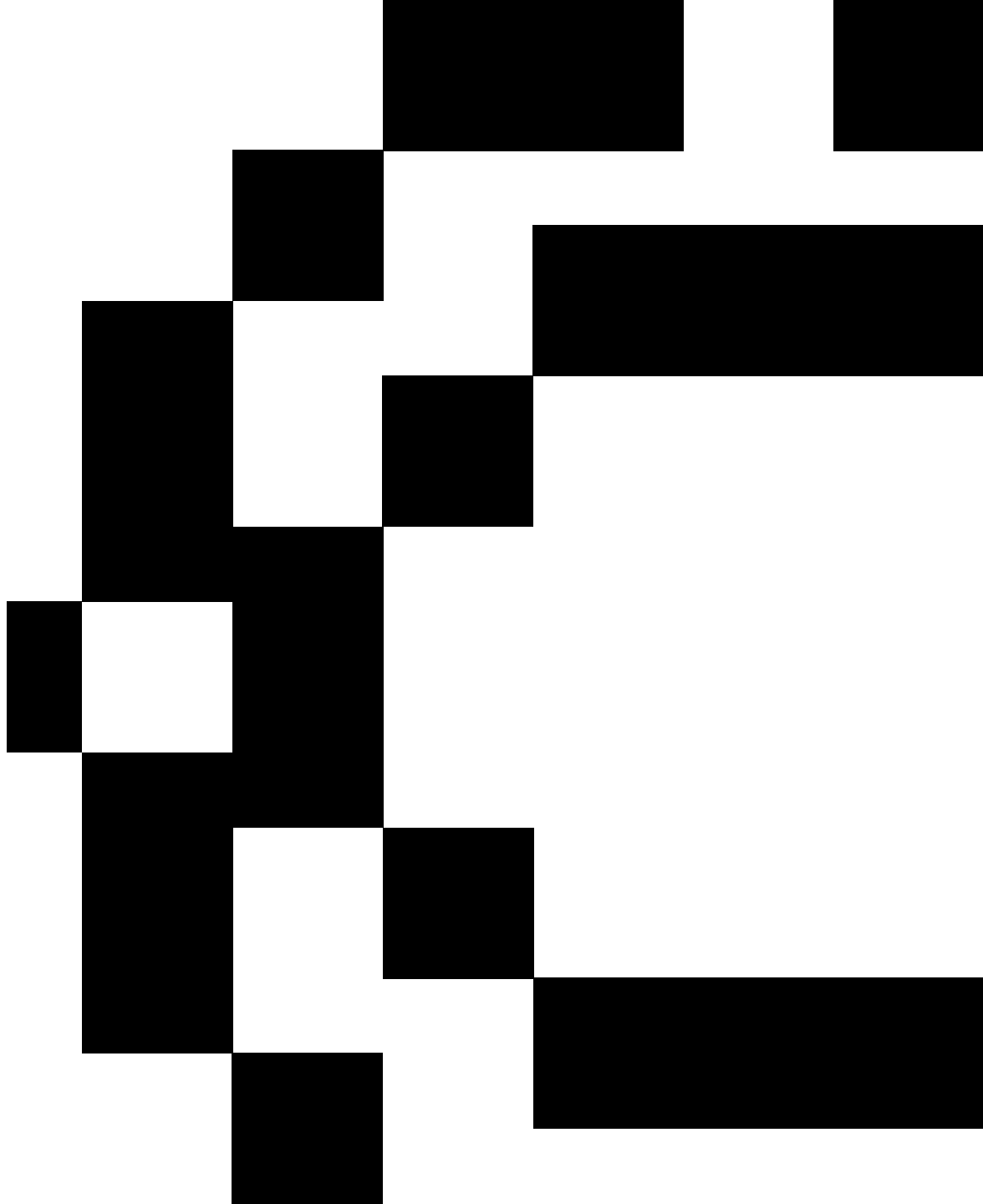
NOVA IMS INFORMATION MANAGEMENT SCHOOL

Computation I / Introduction to Programming

Practicals Week 05 - Conditionals



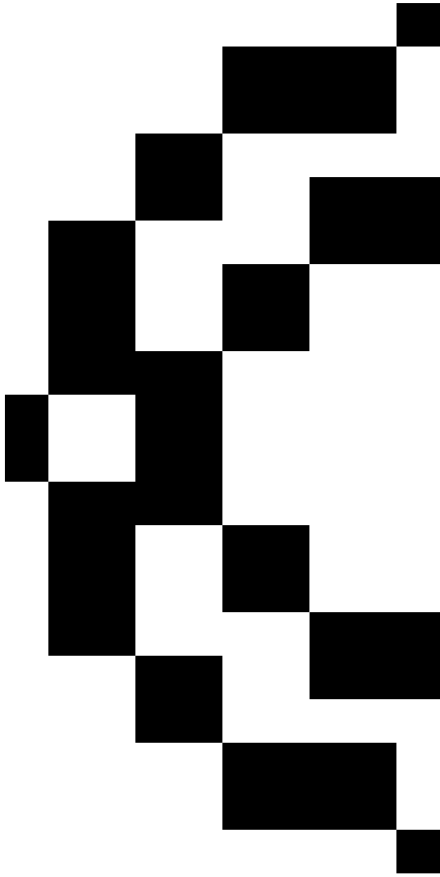
Exercise 1

Ask the user to type their favourite colour. If the colour is yellow, tell them they have terrible taste and you do not understand them. Otherwise, print a sentence saying you can see why that colour would be appealing to them.

Examples

- `Input:  color = "yellow"`
- `Output: You have terrible taste and I do not understand you.`

- `Input:  color = "blue"`
- `Output: I can see why blue would be appealing to you`



Exercise 1 — Solution

```
color = input("Insert your favourite color: ")
if color.lower() == "yellow":
    print("You have terrible taste and I do not understand you.")
else:
    print(f"I can see why {color} would be appealing to you")
```

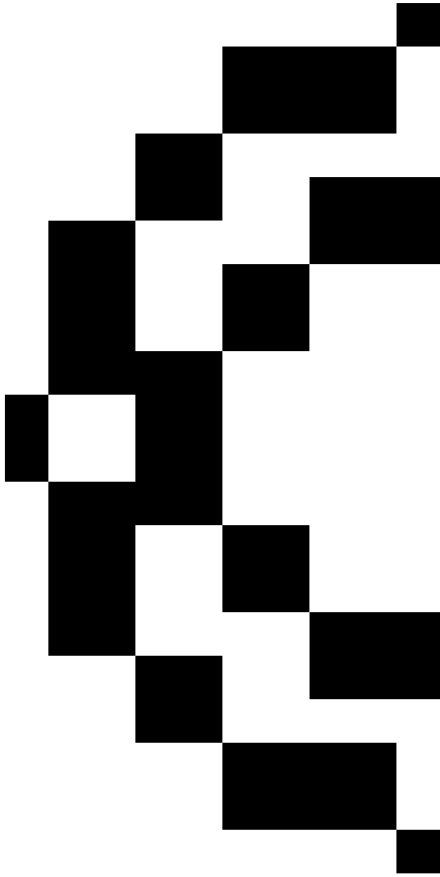
Exercise 2

Write a program that reads an integer from the user. If the number is zero or positive, calculate and print its square root. If the number is negative, print a message saying the number inserted is negative.

Examples

- `Input:  number = 16`
- `Output: The square root of 16 is 4.0`

- `Input:  number = -9`
- `Output: The number is negative`



Exercise 2 — Solution

```
import math

number = int(input("Insert an integer number: "))
if number >= 0:
    print(f"The square root of {number} is {math.sqrt(number)}")
else:
    print("The number is negative")
```

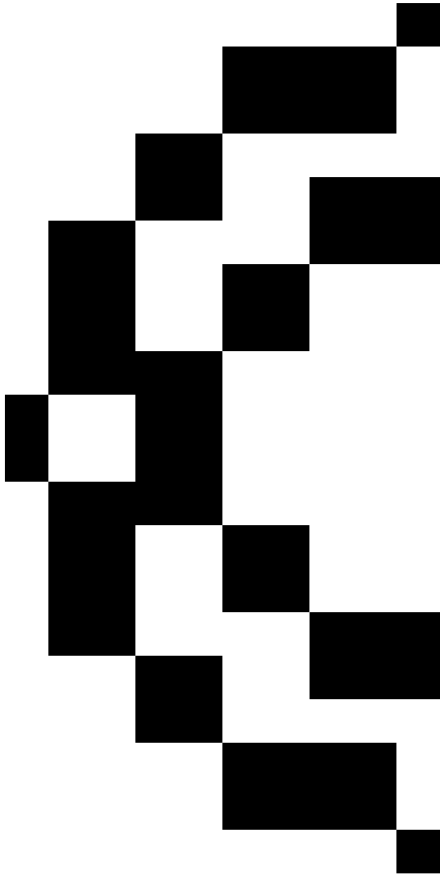
Exercise 3

Ask the user for their name, then ask them (as a Boolean, typed as `True` or `False`) whether milk should be added before the cereal. If they answer `True`, inform them they will not pass Computation I. If they answer `False`, tell them they have some chance of passing. If they type anything else, tell them they did not enter a Boolean value.

Examples

- `Input:  name = "Alex", answer = "True"`
- `Output: Dear Alex, you cannot pass Computation I, good luck with your life.`

- `Input:  name = "Sam", answer = "False"`
- `Output: Dear Sam, congrats, you have chance to pass this class!`



Exercise 3 — Solution

```
name = input("Please insert your name: ")
answer = input("Do you think milk should be added before cereals? ")

if answer == "True":
    print(f"Dear {name}, you cannot pass Computation I, good luck with your life.")
elif answer == "False":
    print(f"Dear {name}, congrats, you have chance to pass this class!")
else:
    print("You did not insert a Boolean value. Try again!")
```

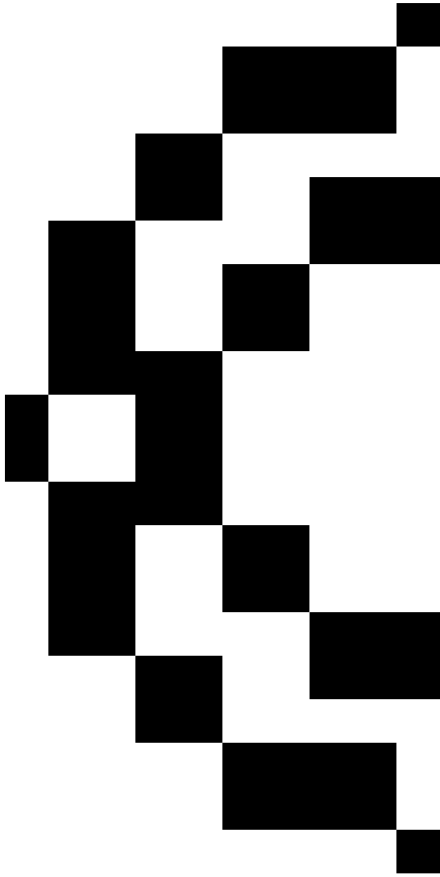
Exercise 4

Ask the user for two numbers, `a` and `b`. Then ask them to choose an operation:
`1` for multiplication, `2` for addition, `3` for `a` raised to the power of `b`.
Print the result of the chosen operation.

Examples

- `Input:  a = 4, b = 2, choice = 1`
- `Output: The result is: 8.0`

- `Input:  a = 4, b = 2, choice = 3`
- `Output: The result is: 16.0`



Exercise 4 — Solution

```
number1 = float(input("Number 1: "))
number2 = float(input("Number 2: "))
expression = int(input("What you want me to do with them? 1: *, 2: +, 3: power "))

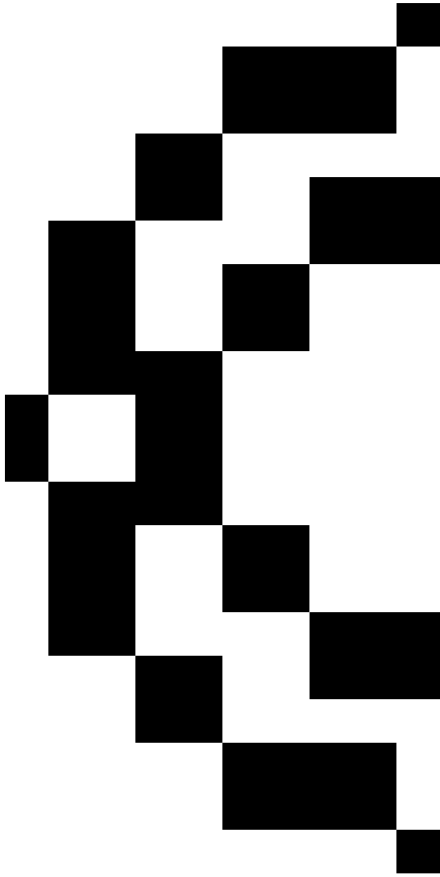
if expression == 1:
    print("The result is:", number1 * number2)
elif expression == 2:
    print("The result is:", number1 + number2)
elif expression == 3:
    print("The result is:", number1 ** number2)
else:
    print("Invalid input")
```

Exercise 5

Gertrude went shopping for soda again. Ask the user how many sodas she bought and how much she paid for each one. Calculate and print the total. If she paid more than €10 in total, tell her she is spending too much; otherwise, tell her she is within budget. Separately, if she bought more than 5 sodas, warn her that she is drinking too many — regardless of what was printed about the budget.

Example

```
Input:  number = 6, price = 2.00
Output: Gertrude spent 12.0 on sodas
        She is spending too much!
        Gertrude is drinking an unhealthy amount of soda...
```



Exercise 5 — Solution

```
number = int(input("How many sodas did she buy? "))
price = float(input("How much did each one cost? "))
total = number * price
print(f"Gertrude spent {total} on sodas")

if total > 10:
    print("She is spending too much!")
else:
    print("She is within budget")

if number > 5:
    print("Gertrude is drinking an unhealthy amount of soda...")
```

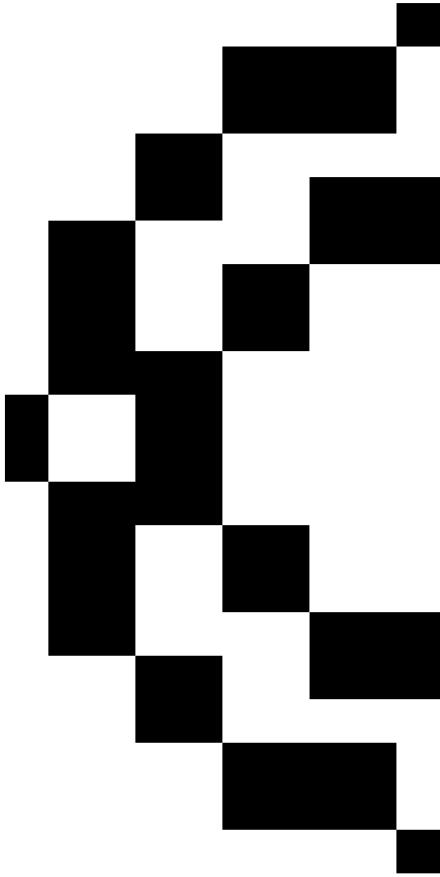
Exercise 6

Ask the user to type an integer. If the number is smaller than 1000, print that the number is small. Otherwise, print how many digits the number contains.

Examples

- `Input:  integer = 42`
- `Output: Your number is small`

- `Input:  integer = 48213`
- `Output: 48213 has 5 digits`



Exercise 6 — Solution

```
integer = int(input("Please insert an integer: "))
if integer < 1000:
    print("Your number is small")
else:
    print(f"{integer} has {len(str(integer))} digits")
```

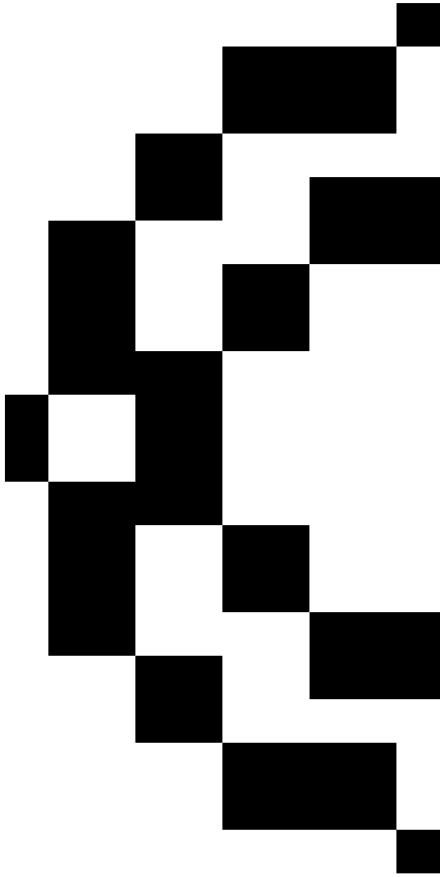
Exercise 7

Ask the user for their phone number, including the country code at the start.
Using the table below, print the name of the country if the code is recognised,
or a message saying the country is not recognised otherwise.
+351 → Portugal
+90 → Turkey
+354 → Iceland

Examples

- `Input:  phone = "+351912345678"`
- `Output: Your number is Portuguese!`

- `Input:  phone = "+90212345678"`
- `Output: Your number is Turkish!`



Exercise 7 — Solution

```
phone = input("Insert your phone number with the country code: ")
if phone[0:4] == "+351":
    print("Your number is Portuguese!")
elif phone[0:3] == "+90":
    print("Your number is Turkish!")
elif phone[0:4] == "+354":
    print("Your number is Icelandic!")
else:
    print("Sorry, I don't recognize your country")
```

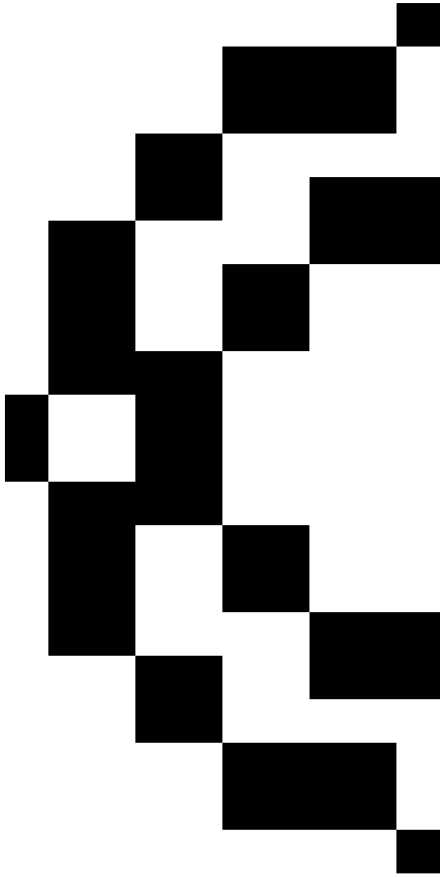
Exercise 8

Ask the user for their name. If it ends in `"a"`, replace the final `"a"` with `"inha"`. If it ends in `"o"`, replace the final `"o"` with `"inho"`. Otherwise, leave the name unchanged. Print a friendly greeting using the (possibly changed) name.

Examples

- `Input:  name = "Carolina"`
- `Output: Nice to meet you Carolininha`

- `Input:  name = "Bruno"`
- `Output: Nice to meet you Bruninho`



Exercise 8 — Solution

```
name = input("Insert your name: ")

if name[-1] == "a":
    name = name[:-1] + "inha"
    print(f"Nice to meet you {name}")
elif name[-1] == "o":
    name = name[:-1] + "inho"
    print(f"Nice to meet you {name}")
else:
    print(f"Nice to meet you {name}")
```

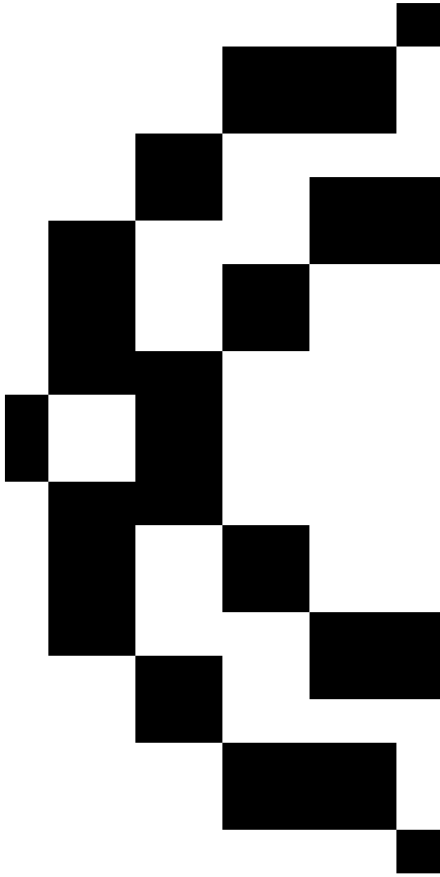
Exercise 9

Ask the user to say something about Nova. If their sentence contains the word "Nova" (in any capitalisation), replace it with "Universidade Nova de Lisboa" and print the updated sentence. Otherwise, ask them if they are sure they are talking about Nova.

Examples

- `Input:  sentence = "I study at Nova and I love it"`
- `Output: i study at universidade nova de lisboa and i love it`

- `Input:  sentence = "I like pizza"`
- `Output: Are you sure you are talking about Nova?`



Exercise 9 — Solution

```
sentence = input("Say something about Nova: ")
sentence = sentence.lower()
if "nova" in sentence:
    print(sentence.replace("nova", "universidade nova de lisboa"))
else:
    print("Are you sure you are talking about Nova?")
```

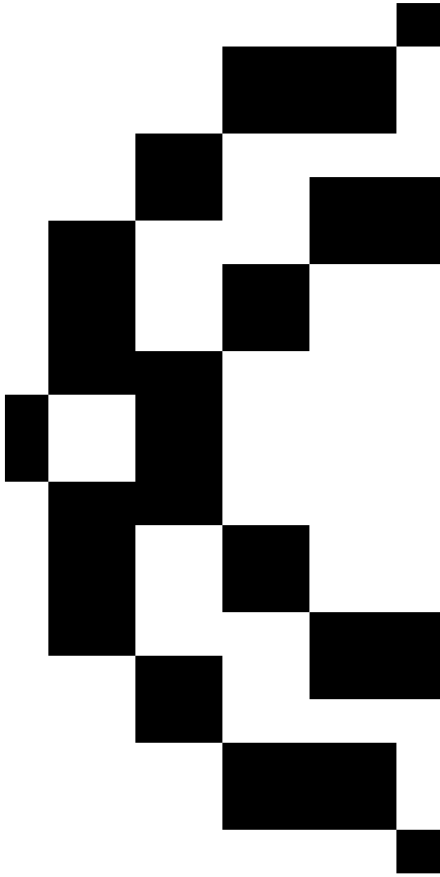
Exercise 10

Ask the user for the name of their pet, then ask them to choose its species:
1 for dog, 2 for cat, 3 for any other animal. Print the cost of a vet
visit: €50 for a dog, €40 for a cat, and €60 for anything else.

Examples

- Input: `pet = "Rex", species = 1`
- Output: `The visit will cost you 50€`

- Input: `pet = "Whiskers", species = 2`
- Output: `The visit will cost you 40€`



Exercise 10 — Solution

```
pet = input("Insert the name of your pet: ")
species = int(input(f"Choose the species of {pet} 1:dog, 2:cat, 3:other "))

if species == 1:
    print("The visit will cost you 50€")
elif species == 2:
    print("The visit will cost you 40€")
elif species == 3:
    print("The visit will cost you 60€")
else:
    print("Invalid input. Try again.")
```

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Thank you

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